Momentum-Enhanced Dual-Prototype Learning Framework for Robust Few-Shot Hyperspectral Image Classification

Hanchi Liu, Jinrong He, *Member, IEEE*, Xiangqing Zhang, and Zhaokui Li, *Member, IEEE*

***Abstract*—Prototypical network-based few-shot learning (FSL) has demonstrated promising performance for hyperspectral image (HSI) classification tasks under scarce sample conditions. However, existing prototype-based FSL methods suffer from data distribution variations among randomly sampled tasks, leading to unstable class prototype representations and weak cross-task generalization with limited samples. To address this issue, we propose a momentum-enhanced dual-prototype learning (MEDPL) framework for robust few-shot HSI classification. First, a momentum-updated prototype mechanism constructs an iteratively optimized prototype memory bank. It obtains accumulated prototypes by exponentially decaying weighted fusion of historical and current prototypes, significantly suppressing noise from randomly sampled data and class center shifts caused by distribution bias. Simultaneously, a class-conditioned perturbation-augmentation strategy is introduced. It generates adaptive noise perturbations for support set features based on learnable covariance matrices to obtain enhanced prototypes, thereby improving the generalization representation capability of class prototypes across tasks. Second, a dual-prototype metric learning framework is designed, jointly utilizing accumulated prototypes and enhanced prototypes to synergistically enhance the model's classification stability and cross-task generalization, thus significantly improving the robustness of few-shot classification. Experimental results demonstrate that MEDPL outperforms other few-shot HSI classification methods. Our source code is available at https://github.com/hejinrong/MEDPL**

***Index Terms*—Few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning, prototypical network.**

I. Introduction

HYPERSPECTRAL images (HSIs) are 3-D data cubes that simultaneously contain rich spatial and spectral information. Compared to traditional-color images, they possess much higher spectral resolution, typically comprising tens to hundreds of contiguous narrow bands [1]. This enables the exploitation of subtle spectral signatures to identify and discriminate various materials and substances. Consequently, hyperspectral imaging has been widely adopted in remote sensing [2], [3], [4], precision agriculture [5], [6], [7], water-quality monitoring [8], [9], food-quality control [10], [11], [12], and military applications [13], [14], [15]. Based on feature extraction and modeling approaches, HSI classification can be categorized into two major classes: traditional methods and deep learning methods. The first category combines hand-crafted features with traditional classifiers, such as k-nearest neighbors (NNs) [16], support vector machines (SVMs) [17], and random forests (RFs) [18]. However, these methods struggle to fully exploit the latent information within hyperspectral data, and the hand-crafted features rely heavily on human prior knowledge, making them inadequate for nonlinear feature representation in complex scenarios. In recent years, deep-learning-based HSI classification has achieved remarkable success, benefiting from its powerful capacity to automatically learn features from high-dimensional data in an end-to-end fashion [19]. Capitalizing on the ability of convolutional neural networks (CNNs) to capture local image patterns, Kanthi et al. [20] proposed a 3-D CNN architecture that jointly extracts spatial–spectral characteristics of HSIs. To mitigate the gradient-vanishing problem inherent in deep CNNs, Zhong et al. [21] introduced a spectral–spatial residual network (SSRN) that employs spectral and spatial residual blocks to learn discriminative features from the abundant spectral and spatial information. Zhang et al. [22] further presented a context-aware CNN framework based on diversified region inputs, constructing a joint representation by fusing multiscale spectral–spatial features and semantic contextual information. However, CNN-based methods are still limited in fully exploiting the sequential nature of spectral features. To address this, Yang et al. [23] proposed the HSI transformer (HiT), which integrates convolution into the transformer architecture. By designing a spectral-adaptive 3-D convolutional projection module and a convolutional permutation module, HiT effectively captures subtle differences in spectral sequences as well as local spatial contexts. To alleviate the high computational complexity of transformers, Zhang et al. [24] introduced a lightweight transformer network (LiT), in which channel-wise and position-wise lightweight self-attention modules significantly reduce computational overhead. Nonetheless, all these methods typically require a large amount of labeled data for training. Because labeling HSIs is extremely costly, their performance drops sharply in scenarios where annotated samples are scarce.

To tackle the scarcity of labeled samples, few-shot learning (FSL) has been applied to HSI classification tasks. FSL aims to enable rapid learning and generalization to novel

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Hanchi Liu, Jinrong He, and Xiangqing Zhang are with the College of Mathematics and Computer Science, Yan'an University, Yan'an 716000, China (e-mail: liuhanchi@yau.edu.cn; hejinrong@yau.edu.cn; zhangxiangqing@yau.edu.cn).

Zhaokui Li is with the School of Computer Science, Shenyang Aerospace University, Shenyang 110136, China (e-mail: lzk@sau.edu.cn).

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classes using only a limited number of samples [25]. Current FSL methods can be categorized into four major types: data augmentation [26], meta-learning [27], fine-tuning [28], and metric learning [29]. These approaches address the challenges of few-shot HSI classification from different perspectives: sample quantity, learning paradigms, model parameters, and feature representation, respectively. Data augmentation methods learn augmentation techniques on classes with abundant samples and apply them to few-shot classes to expand data. Wei et al. [30] proposed a relation-based generative adversarial network (FSHyperRGAN), which uses GANs to generate virtual samples for training data expansion, combined with a relational feature extraction module to guide class-specific sample generation. However, such methods often suffer from low generation quality and diversity under few-shot conditions and may introduce irrelevant noise. Meta-learning methods learn optimization strategies and initializations from training data of classes with abundant samples to rapidly adapt to few-shot tasks. Zhang et al. [31] proposed a Bayesian meta-learning approach for few-shot classification, learning hierarchical probabilistic priors through meta-tasks, modeling the posterior distribution of prototype vectors, and leveraging variational inference to mitigate few-shot estimation bias. Nevertheless, this method exhibits high sensitivity to task distribution—performance plummets if test tasks deviate from the meta-training distribution. Fine-tuning methods first train a model with strong generalization capabilities on a large-scale dataset, then fine-tune it on the few-shot dataset. Wang et al. [32] proposed a heterogeneous FSL (HFSL) framework, which fine-tunes pretrained natural image models on hyperspectral data, achieving optimal classification with minimal annotations. However, fine-tuning still shows limitations in few-shot scenarios, where domain differences can cause negative transfer. Metric learning methods include Siamese networks, relation networks, and prototype networks. They learn sample similarities using data from classes with abundant samples and then classify samples from few-shot classes. Xue et al. [33] introduced a spectral–spatial Siamese network (S3Net), which employs a dual-branch lightweight network fed with differentiated sample pairs to enhance feature representation. Gao et al. [34] utilized a relation network model, implementing relational modeling between samples through a learnable nonlinear similarity function. While Siamese and relation networks improve feature discriminability, they become computationally inefficient for complex HSI data due to sample pair construction and pairwise similarity calculations. These limitations have motivated the adoption of a prototypical network, which significantly reduce complexity by computing distances from class centers.

In prototypical networks, class prototypes are computed as the mean embeddings of the support samples of each class in an embedding space, and classification is performed by assigning each query sample to the nearest class prototype in terms of Euclidean distance. Liu et al. [35] introduced a deep FSL (DFSL) approach that employs a residual 3-D CNN to extract spectral–spatial features and uses meta-task training to learn a metric space for class-prototype-based classification. Li et al. [36] proposed deep cross-domain FSL (DCFSL), which mitigates domain shift via conditional adversarial domain adaptation. Zhang et al. [37] presented a graph information aggregation cross-domain FSL (Gia-CFSL) framework that complements prior work by integrating prototype learning with graph-based information aggregation to exploit nonlocal spatial cues. Building on conditional adversarial training, Ye et al. [38] introduced the adaptive domain-adversarial FSL (ADAFSL) framework, which dynamically modulates the conditional adversarial loss through a class-difference-aware weighting mechanism. Qin et al. [39] developed a feature-disentangled cross-domain FSL (FDFSL) method that, within the prototype-metric classification paradigm, suppresses source-domain bias via orthogonal low-rank feature decomposition and prevents over-fitting in domain alignment through a self-distillation strategy. Li et al. [40] proposed a semantic-guided prototypical learning scheme that augments prototype expressiveness by one-hot encoding class labels and employing a prototype generator to produce more discriminative prototype features. Meanwhile, Liu et al. [41] introduced multilevel prototype alignment (MLPA), which enforces feature alignment at multiple layers of the feature extractor and couples it with class-prototype-based domain-adversarial training to enhance generalization and classification accuracy on the target domain. Across all these methods, the quality of prototype representation is crucial to classification performance. However, the single mean prototype estimated from scarce data is often unstable and sensitive to distribution shifts; the model can easily over-fit to specific noise or biases present in the few samples.

To address the instability and weak generalization of single-mean prototypes in metric-based few-shot HSI classification, we present a novel momentum-enhanced dual-prototype learning (MEDPL) framework. First, a momentum-updated module accumulates historical prototype information to stabilize class centers, while a perturbation-augmentation module injects adaptive class-conditional noise to improve the generalizability of prototype representations. Second, a dual-prototype metric-learning mechanism jointly optimizes the model, significantly enhancing classification robustness under extreme data scarcity.

The main contributions of this article are as follows.

1) We propose a novel dual-prototype learning framework for robust few-shot HSI classification. By jointly leveraging accumulated and enhanced prototypes in metric learning, the model significantly improves both discriminative power and generalization across tasks with diverse data distributions.
2) A momentum update mechanism was designed, accumulating historical class prototype information via an exponential moving average. It enhances prototype stability and effectiveness rather than interclass discriminability, yielding more representative prototypes and reducing bias from random sampling and noisy samples.
3) A perturbation-augmentation strategy is introduced that injects class-conditional noise into the feature space, perturbing support set features to obtain more generalizable enhanced prototypes. This boosts the model's generalization for N-way classification tasks and substantially increases robustness under few-shot scenarios.

II. Related Work

A. Few-Shot Learning

FSL [42] aims to address the challenge of model generalization in scenarios with extremely scarce labeled data. Formally, it can be defined as a meta-learning problem: the goal is to enable the model to learn a learning algorithm

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itself, allowing it to quickly adapt its parameters or decision mechanisms to new tasks with only a small number of target task samples [43]. To achieve this, the model is first pretrained on source tasks or large datasets with abundant labeled data to learn general feature representations or optimization strategies. Subsequently, the acquired knowledge is transferred to the target task, where only a minimal amount of labeled data from the target task is required for fine-tuning or direct application, enabling efficient adaptation to the new task. FSL primarily includes methods based on meta-learning and metric learning.

Meta-learning-based methods enhance generalization under data scarcity by accumulating cross-task knowledge through multitask training. Rusu et al. [44] proposed latent embedding optimization (LEO), which performs gradient-based meta-learning in a low-dimensional latent space to generate and adapt model parameters, enabling efficient adaptation in few-shot tasks. Jia et al. [45] introduced Mix-MAML, a hybrid optimization meta-learning method that combines data augmentation, initialization decay, and resolution increase to improve the generalization performance of model-agnostic meta-learning in few-shot classification.

Metric-based methods first map data into a low-dimensional metric space, then perform classification based on distances between embedded features. Lim et al. [46] proposed a three-stage training framework (SSL-ProtoNet) combining self-supervised learning, prototypical networks, and knowledge distillation, which significantly improved few-shot image classification performance on multiple benchmark datasets by enhancing sample discriminability. Palanisamy et al. [47] introduced the Proto-CLIP framework, which enhances few-shot classification performance by jointly aligning image and text prototypes of the CLIP model.

B. Prototypical Network

Prototypical networks [48] operate on the core principle of constructing a representative prototype vector for each class, which characterizes the distribution center of that class's samples in the embedding space. Prototypical networks typically adopt episodic training as the meta-learning paradigm to faithfully mimic few-shot scenarios [49], [50]. In this paradigm, each episode is constructed as an independent "mini-task," i.e., an N-way K-shot problem: N classes are randomly selected from the dataset; for each of these classes, K labeled samples are drawn to form the support set, while additional unlabeled samples from the same classes constitute the query set. Within each episode, the model computes the N class prototypes by averaging the embeddings of the support samples and then assigns each query sample to the class whose prototype exhibits the smallest Euclidean distance. The support set labels provide the supervisory context for building the classifier, while the query set labels are used to compute loss and optimize the model, together forming the supervised learning basis of the approach.

In task-based FSL, the value of K in each task is typically very small. Due to the randomness of sample selection, the support set may contain samples that are far from the class centroid, causing prototypes obtained through simple feature averaging to be unrepresentative. To address this issue, Xue and Wang [51] proposed the RestoreNet model, which significantly improves the accuracy of one-shot image classification by learning to restore features of noisy images toward class centroids. Zhu and Koniusz [52] introduced a prototype-based label propagation method (protoLP) that enhances classification performance in transductive FSL by iteratively optimizing the bipartite graph structure between prototypes and samples.

III. Methodology

Under the few-shot setting, we have a source domain dataset D_S containing C_s classes and a target domain dataset D_T containing C_t classes, where $C_s > C_t$. The target domain dataset D_T is divided into a labeled few-shot dataset D_f and an unlabeled test dataset D_t, with $D_f \cap D_t = \emptyset$. Under the episodic training paradigm, exemplified using the source domain, an N-way K-shot task consists of a support set $S_s = \{(x_i, y_i)\}_{i=1}^{N\times K}$ and a query set $Q_s = \{(x_j, y_j)\}_{j=1}^{N\times Q}$. Finally, $f_\theta(\cdot)$ represents a feature extractor parameterized by θ.

A. Overall Architecture

The framework of the proposed method is illustrated in Fig. 1. It comprises four modules: the mapping layer and feature extractor module, the momentum update module, the perturbation augmentation module, and the dual-prototype metric learning module. Initially, owing to the limited sample size of the few-shot dataset, which obstructs the construction of sufficient meta-tasks, random Gaussian noise is introduced to augment the labeled target-domain samples. Subsequently, samples from both the source and target domains pass through their respective mapping layers to ensure consistent input dimensions, after which a shared-parameter feature extractor embeds them into a unified spectral–spatial latent space. Thereafter, to derive more representative class prototypes, support-set features from both domains are processed by the momentum-updating module, which incrementally aggregates historical class-prototype information to yield accumulated prototypes. Concurrently, to enhance the generalizability of each prototype, the identical support-set features are subjected to the perturbation-augmentation module, wherein adaptive class-conditional noise is injected to generate enhanced prototypes. Ultimately, dual-prototype metric learning is conducted across the source and target domains to refine the parameters of the shared feature extractor. Consequently, this joint optimization not only strengthens the model's discriminative capacity but also improves its generalization performance, thereby facilitating the transfer of generic discriminative knowledge from the source domain to the target domain.

B. Mapping Layer and Feature Extractor

The hyperspectral cube of each sample from both source and target domains is encoded into spatial–spectral embedded features through domain-specific mapping layers and a shared feature extractor. Each mapping layer comprises d 2-D convolutional kernels of size $1 \times 1 \times$ ch, where ch denotes the input channel count and d represents the output channel dimension. This configuration maps data from different input dimensionalities across domains into an identical d-dimensional channel representation.

The feature extractor is illustrated in Fig. 2. It consists of spatial and spectral extraction branches that, respectively, capture the neighborhood spatial and spectral information of the center pixel. In the spatial branch, the reduced dimension sample first passes through a 3-D convolution layer with 24 kernels of size $1\times1\times100$, compressing the spectral dimension to one channel while keeping spatial resolution so the focus falls on spatial cues. A residual block of two identical 2-D

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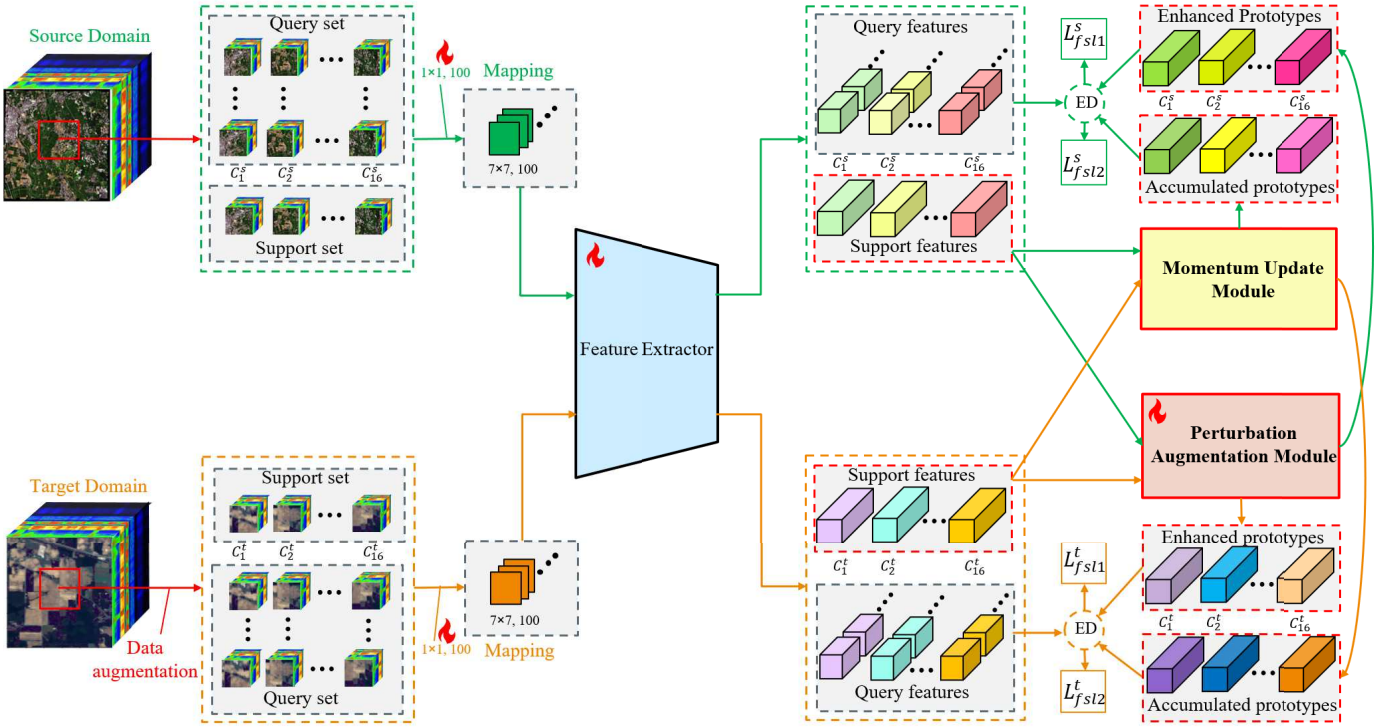


Fig. 1. Framework of proposed MEDPL in the training phase. It adopts an episodic training paradigm on both the source and target domains to conduct meta-learning. After feature extraction of the support set samples, more class-representative accumulated prototypes and more robust enhanced prototypes are generated. Finally, dual-prototype metric learning is performed.

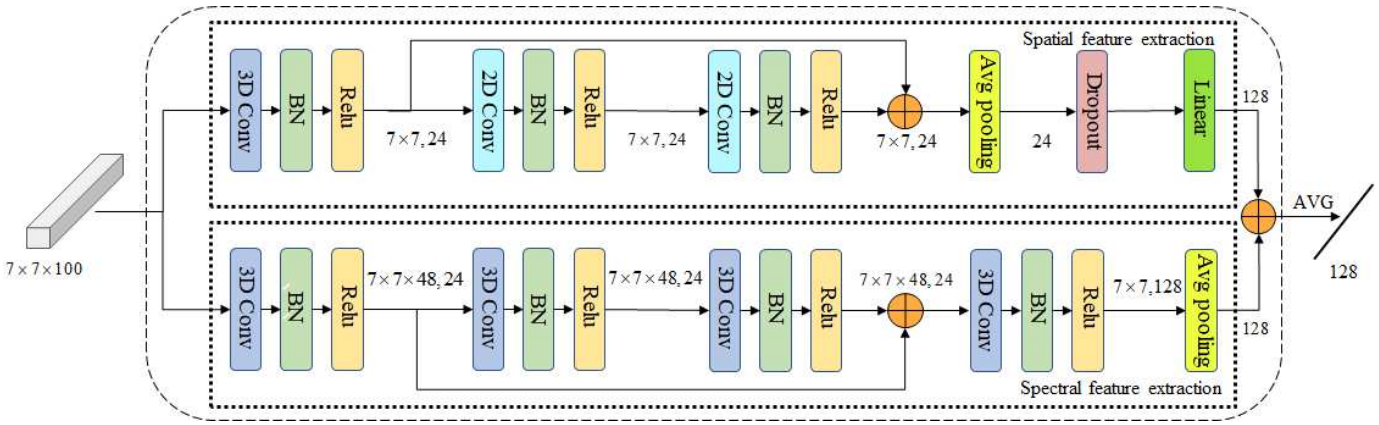


Fig. 2. Architecture of feature extractor. The spatial feature extraction branch extracts spatial structure information, the spectral feature extraction branch extracts spectral sequence information, and finally, they are equally weighted and fused into a spatial–spectral joint feature.

convolution layers, each with 24 kernels of size 3×3, then refines these spatial cues. Global average pooling yields a spatial feature vector, whose dimension is aligned with that of the spectral vector via a linear layer. In the spectral branch, the sample is first processed by a 3-D convolution layer with 24 kernels of size $1 \times 1 \times 7$ to extract local band correlations along the spectral axis. A residual block of two identical 3-D convolution layers, each with 24 kernels of size $1 \times 1 \times 7$, further enhances spectral representation. A final 3-D convolution layer with 128 kernels of size $1 \times 1 \times 48$ adjusts the channel dimension, and global average pooling produces a spectral feature vector matching the spatial one. The complementary spatial and spectral vectors are then fused by equal-weighted addition to form a joint spatial spectral feature that further improves classification performance [53].

C. Momentum Update Module

In FSL, the prototype within the current episode represents the representative feature of each class. It is obtained by averaging the features of the samples belonging to that class, thereby establishing a decision boundary for classification. For class c, its class prototype is calculated as follows:

$$P_c = \frac{1}{k} \sum_{x_i \in W_c} f_\theta(x_i) \tag{1}$$

where W_c denotes the set of k labeled samples of class c in the current support set, and $f_\theta(x_i)$ is the feature vector of the ith sample extracted by the feature extractor.

As shown in (1), the limited samples randomly sampled across different tasks exhibit distributional discrepancies and

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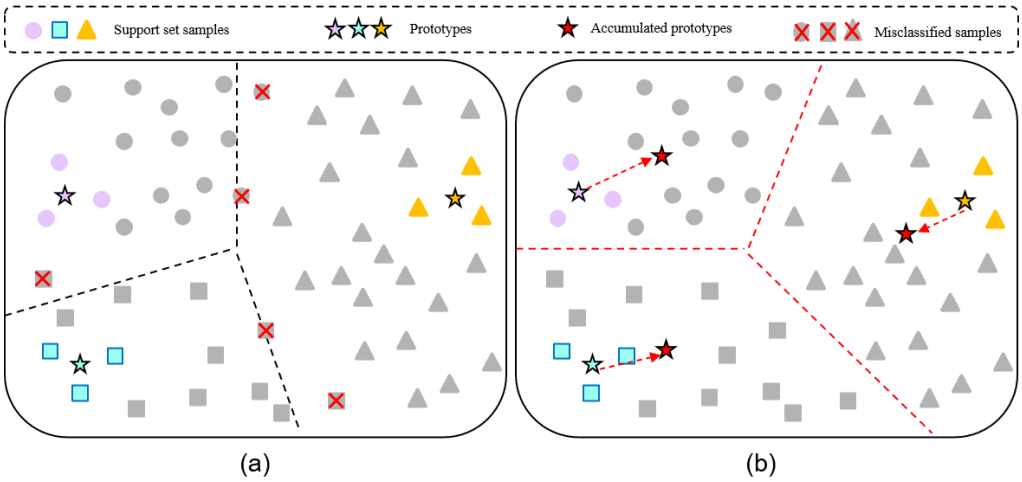


Fig. 3. Illustration of prototype stability and effectiveness. (a) In mean-based prototype methods, due to the distribution bias between randomly selected support set samples, the prototypes computed in different episodes may deviate from the true class center, easily leading to misclassification of samples near class boundaries. (b) Our method enhances the effectiveness of class prototypes by leveraging global prototype information, making the accumulated prototypes closer to the true class center, thereby improving classification accuracy.

susceptibility to noisy samples. This causes the computed prototypes to deviate from the true class prototypes of the entire sample population, as illustrated in the left of Fig. 3. Consequently, the instability of classification decision boundaries leads to frequent misclassifications of samples near the true class boundaries. To address this issue, inspired by [54], we propose a momentum-updating mechanism that enhances the effectiveness of prototypes through a historical prototype memory bank. Specifically, we construct a source domain prototype memory bank $M_s = \{p_c^{\text{acc}}\}_{c=1}^{N}$ and a target domain prototype memory bank $M_t = \{p_c^{\text{acc}}\}_{c=1}^{N}$, which are continuously updated during training. Historical prototypes are initialized as zero vectors to eliminate initial bias and iteratively updated to preserve long-term statistical characteristics of class prototype features. For class c in the prototype memory bank, its updated accumulated prototype is calculated as follows:

$$p_c^{\text{acc}} = \gamma \cdot p_c^{\text{acc}} + (1-\gamma) \cdot p_c^{\text{new}} \tag{2}$$

where p_c^{new} represents the new prototype computed for the current task, and $\gamma \in [0, 1]$ denotes the momentum coefficient. By weighted fusion of historical prototypes and the current prototype, interference from noisy samples is exponentially decayed. Simultaneously, leveraging the global prototype statistics of the overall sample distribution enables the learned prototypes to progressively converge toward the true class centers and stabilize with increasing iteration rounds, as visualized in the right of Fig. 3. This mechanism significantly enhances classification capability.

D. Perturbation Augmentation Module

In cross-domain few-shot scenarios, limited target domain samples are prone to cause overfitting issues [55]. To enhance the model's generalization capability for N-way classification, we propose a class-conditioned perturbation-augmentation strategy. This approach augments support set features by injecting noise that simulates reasonable intraclass variations rather than disruptive noise into the feature space, yielding more generalizable prototype representations. Simultaneously, it pulls intraclass samples closer to the generalized prototypes

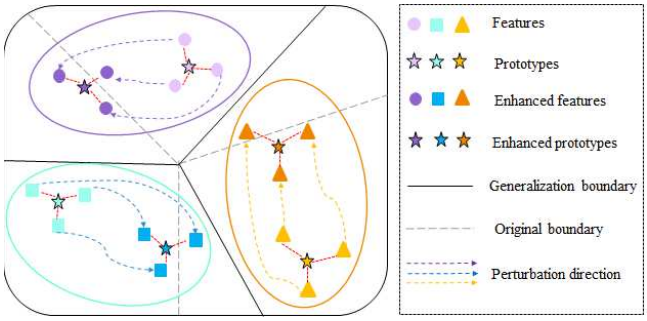


Fig. 4. Schematic of enhanced prototypes. Adaptive noise conforming to class-conditional feature distributions is generated based on the covariance matrix and injected into the features of support set samples, thereby producing enhanced prototypes with greater robustness and generalization capability. This process effectively improves prototype adaptability in few-shot scenarios by simulating intraclass feature diversity.

within the feature space, forcing the model to learn transferable knowledge, as illustrated in Fig. 4.

Specifically, for the embedded feature set $z_c, z_c \in \mathrm{R}^{k\times d}$ of class c in the support set during the current iteration, it can be enhanced as follows:

$$\hat{z}_c = z_c + \eta \cdot \xi_c \cdot 1^{\mathrm{T}}, \quad \xi_c \sim \mathcal{N}\left(0, \sum_c\right) \tag{3}$$

where k denotes the number of samples in the class, d represents the feature dimensionality, $\xi_c \in \mathrm{R}^d$ is the noise vector, $1 \in \mathrm{R}^k$ serves as an all-one vector ensuring features within the same class share identical noise perturbations, and η is a nonnegative scalar hyperparameter controlling the noise intensity. The vector ξ_c is sampled from a class-specific multivariate normal distribution $\mathcal{N}\left(0, \sum_c\right), \sum_c \in \mathrm{R}^{d\times d}$, meaning the perturbation direction aligns with the natural dispersion direction of intraclass features. This ensures that the enhanced prototype representation shifts along relevant directions of the intraclass feature distribution while preserving its class semantic information. Notably, compared to sampling noise from an isotropic standard Gaussian distribution $\mathcal{N}(0, \sigma^2 I)$, our method possesses inherent advantages. The covariance matrix of standard Gaussian sampling is a diagonal matrix

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$\sigma^2 I$, which generates perturbations with uniform intensity and completely random directions across all dimensions of the feature space. Such perturbations often lack specificity and may introduce noise in directions irrelevant to class discrimination, resulting in ineffective enhancement or even interference with feature representation.

Next, estimating class-specific covariance matrices poses a key challenge. For the feature set $z_c = \{z_c^i \in \mathrm{R}^d\}_{i=1}^k$ of class c, the unbiased estimate of its covariance matrix should be calculated as

$$\sum_c = \frac{1}{k-1} \sum_{i=1}^{k} \left(z_c^i - \mu\right) \left(z_c^i - \mu\right)^{\mathrm{T}} \tag{4}$$

where $\mu = 1/k \sum_{i=1}^{k} z_c^i$ denotes the mean vector of the feature set z_c. Equation (4) reveals that under few-shot conditions where $k \ll d$, covariance estimation error escalates drastically, resulting in unreliable covariance matrices. To effectively model interfeature dimensional correlations, we construct a set of N learnable covariance matrices $\left\{\sum_c \in \mathrm{R}^{d\times d}\right\}_{c=1}^N$ as model parameters for each N-way task. These matrices undergo end-to-end data-adaptive training. At training initialization, the covariance matrices are set to

$$\sum_c = 0.1I \tag{5}$$

where $I \in \mathrm{R}^{d\times d}$ denotes the identity matrix, $c \in [1, N]$. During the initial training phase, when the model has not yet fully learned the distribution structure of the feature space and correlations between feature dimensions are weak, employing a diagonal matrix for initialization helps avoid introducing erroneous prior correlations while maintaining numerical stability. Furthermore, the small initial variance prevents excessive perturbation magnitude, thereby preserving intraclass semantic consistency. As parameter optimization proceeds through backpropagation, the model will gradually learn the characteristic distribution patterns within different classes.

Finally, the enhanced prototype after perturbation is obtained as

$$p_c^{\mathrm{enh}} = \frac{1}{k} \sum_{i=1}^{k} \hat{z}_c^i. \tag{6}$$

E. Dual Prototype Metric Learning

Conventional FSL based on a single prototype performs classification by computing distances between the query set embeddings and each class prototype within the spatial–spectral embedding space. However, this single-prototype metric learning mechanism struggles to simultaneously achieve high classification accuracy on the current task and maintain generalization capability across other tasks. To address this limitation, we propose a dual-prototype metric learning framework leveraging both accumulated prototypes and enhanced prototypes. This approach significantly boosts classification accuracy while substantially improving model generalization. The entire training phase is shown in Algorithm 1.

Specifically, during episodic training on the source domain, each task randomly selects N classes, sampling $K + Q$ hyperspectral cubes of size $7 \times 7\times$ ch1 per class as model inputs. The support set serves as labeled samples for computing dual prototypes per class, while the query set acts as unlabeled samples for classification evaluation. Samples are mapped into the feature space via mapping layers and the feature extractor. The model then computes two independent predictive probabilities for each query sample using distance-based softmax activation: discriminative probability P_{dis} measured against accumulated prototypes and generalizability probability P_{gen} measured against enhanced prototypes

Algorithm 1 Training Phase of MEDPL

Input:

The source dataset D_s
The target dataset D_t
The support set S_s and query set Q_s from D_s
The support set S_t and query set Q_t from D_t
Number of training episodes E

Output:

The total loss L_{total} of the model

1. **for** episode =0: E-1 **do**
2. Train with S_s, Q_s, S_t,and Q_t
3. Calculate features z_s, z_t and prototypes p_s, p_t
4. Calculate $p_s \rightarrow p_s^{acc}$ and $p_t \rightarrow p_t^{acc}$ by Eq.(2)
5. Calculate $z_s \rightarrow p_s^{enh}$ and $z_t \rightarrow p_t^{enh}$ by Eq.(3) (6)
6. Calculate the P_{dis} and P_{gen} by Eq.(7) and Eq.(8)
7. Calculate the L_{fsl}^s of an episode by Ep. (9)
8. Calculate the L_{fsl}^t of an episode same as L_{fsl}^s
9. Return the L_{total} by Ep. (10)
10. **end for**

$$P_{\mathrm{dis}}\left(y_j = c | x_j \in Q_s\right) = \frac{\exp\left(-\mathrm{ED}\left(f\left(x_j\right), p_c^{\mathrm{acc}}\right)\right)}{\sum_c^N \exp\left(-\mathrm{ED}\left(f\left(x_j\right), p_c^{\mathrm{acc}}\right)\right)} \tag{7}$$

$$P_{\mathrm{gen}}\left(y_j = c | x_j \in Q_s\right) = \frac{\exp\left(-\mathrm{ED}\left(f\left(x_j\right), p_c^{\mathrm{enh}}\right)\right)}{\sum_c^N \exp\left(-\mathrm{ED}\left(f\left(x_j\right), p_c^{\mathrm{enh}}\right)\right)} \tag{8}$$

where ED(,) denotes the Euclidean distance function, x_j represents the jth query set sample, y_j is its predicted label, and N indicates the number of classes in an episode task. Consequently, the total FSL loss for the source domain is formulated as

$$L_{\mathrm{fsl}}^s = \lambda E_{S_s, Q_s}\left[-\sum_{(x,y)\in Q_s} \log p_{\mathrm{dis}}\left(y = c|x\right)\right] + (1-\lambda) E_{S_s, Q_s}\left[-\sum_{(x,y)\in Q_s} \log p_{\mathrm{gen}}\left(y = c|x\right)\right] \tag{9}$$

where S_s and Q_s denote the support set and query set in the source domain dataset, respectively, while λ controls the loss weight for dual prototypes. Similarly, the FSL loss L_{fsl}^t for the target domain aligns with the formulation of the source domain FSL loss. Overall, the optimization objective of this loss function is to cooperatively maximize the log-likelihood of correct classification for query samples through the dual-prototype mechanism. This drives the model to optimize two key aspects simultaneously: the discriminative probability term based on accumulated prototypes forces query sample features to align closer to their correct classes in the embedding space, thereby improving intraclass compactness and classification accuracy; meanwhile, the generalization probability term based on enhanced prototypes introduces class-conditional perturbations

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TABLE I
ALL DETAILS OF THE EXPERIMENTAL DATASETS

Datasets	Source Domain	Target Domain			
	CH	IP	SA	UP	HT
Spatial size	2517 × 2335	145×145	512 × 217	610×340	349 × 1905
No. of bands	128	102	204	103	144
Spectral range	363–1018	400-2500	400-2500	430-860	380–1050
Spatial Resolution	2.5 m/pixel	20 m/pixel	3.7 m/pixel	1.3 m/pixel	2.5 m/pixel
Total classes	19	16	16	9	15
Labeled samples	77592	10249	54129	42776	15029

TABLE II
SAMPLE COUNTS PER CLASS IN SOURCE AND TARGET DOMAINS

Class	CH		IP		UP		SA		HT	
	Name	Count	Name	Count	Name	Count	Name	Count	Name	Count
1	Water	2345	Alfalfa	46	Asphalt	6631	Brocoli green weeds 1	2009	Healthy grass	1251
2	Bare soil (school)	2859	Corn notill	1428	Meadows	18649	Brocoli green weeds 2	3726	Stressed grass	1254
3	Bare soil (park)	236	Corn mintill	830	Gravel	2099	Fallow	1976	Synthetic grass	697
4	Bare soil (farmland)	48525	Corn	237	Tress	3064	Fallow rough plow	1394	Trees	1244
5	Natural plants	4297	Grass pasture	483	Sheets	1345	Fallow smooth	2678	Soil	1242
6	Weeds in farmland	1108	Grass trees in farmland	730	Bare soil	5029	Stubble	3959	Water	325
7	Forest	20516	Grass pasture mowed	28	Bitumen	1330	Celery	3579	Residential	1268
8	Grass	6515	Hay windrowed	478	Bricks	3682	Grapes untrained	11271	Commercial	1244
9	Rice field (groim)	13369	Oats	20	Shadow	947	Soil vinyard develop	6203	Road	1252
10	Rice field (first stage)	1268	Soybean notill	972			Corn senesced green weeds	3278	Highway	1227
11	Row crops	5961	Soybean mintill	2455			Lettuce romaine 4wk	1068	Railway	1235
12	Plastic house	2193	Soybean clean	593			Lettuce romaine 5wk	1927	Parking Lot 1	1233
13	Manmade (nan-dark)	1220	Wheat	205			Lettuce romaine 6wk	916	Parking Lot 2	469
14	Manmade (dark)	7664	Woods	1265			Lettuce romaine 7wk	1070	Tennis Court	428
15	Manmade (blue)	431	Buildings Grass Trees Drives	386			Vinyard untrained	7268	Running Track	660
16	Manmade (red)	222	Stone Steel Towers	93			Vinyard vertical trellis	1807		
17	Manmade grass	1040								
18	Asphalt	801								
19	Paved ground	145								
Total		77592		10249		42776		54129		15029

to simulate feature variations in real scenarios, compelling the model to learn representations robust to minor variations and enhancing cross-task adaptability.

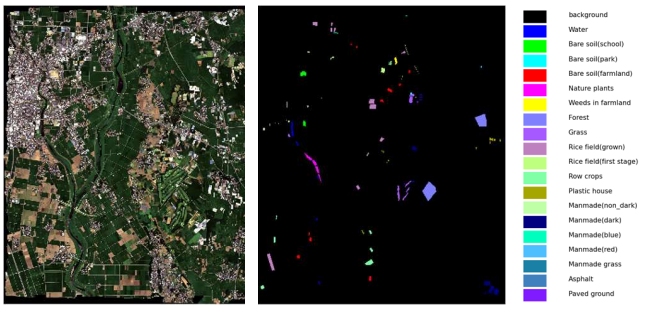
The final loss function of MEDPL is expressed as

$$L_{\text{total}} = L^{s}_{\text{fsl}} + L^{t}_{\text{fsl}}. \tag{10}$$

IV. EXPERIMENTS

A. Data Description

To ensure a fair evaluation of the proposed MEDPL method, we conduct experiments on five publicly available hyperspectral datasets: Chikusei (CH) [56], Indian Pines (IP), University of Pavia (UP), Salinas (SA), and Houston (HT). In our experimental design, CH serves as the source domain to aid model training, while the remaining four datasets are used as target domains to comprehensively assess performance. Detailed acquisition information for each dataset—including spatial dimensions, number of spectral bands, spectral range, spatial resolution, total number of classes, and labeled sample counts—is provided in Table I. Furthermore, per-class sample statistics are summarized in Table II, offering a clear view of the label distribution within each dataset. For intuitive visualization, false-color composites and their corresponding ground truth maps for all datasets are presented in Figs. 5–9.



(a) (b) (c)

Fig. 5. CH dataset: (a) pseudo-color image, (b) ground-truth map, and (c) category of CH.

B. Experimental Setup

Experiments were conducted on a single RTX 4090 GPU with 24-GB memory under Ubuntu 20.04. We used PyTorch 1.12.1, CUDA 11.3, Python 3.8, and optimized parameters

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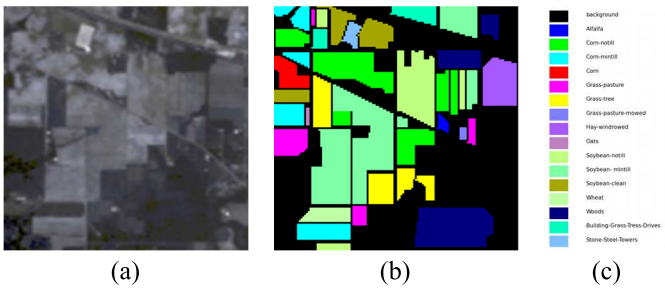


Fig. 6. IP dataset: (a) pseudo-color image, (b) ground-truth map, and (c) category of IP.

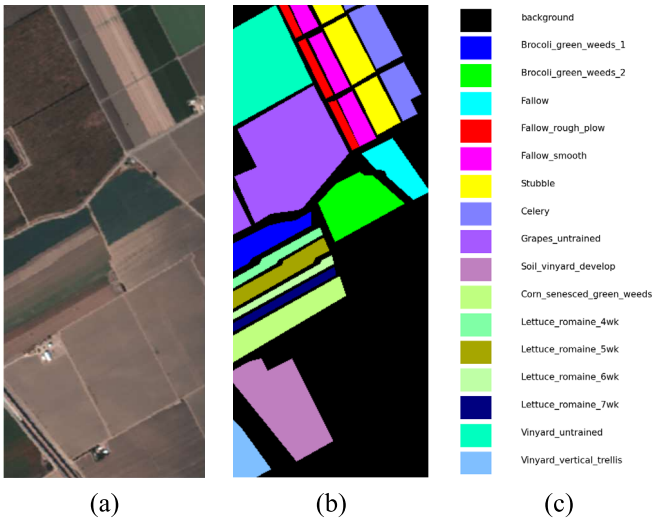


Fig. 7. SA dataset: (a) pseudo-color image, (b) ground-truth map, and (c) category of SA.

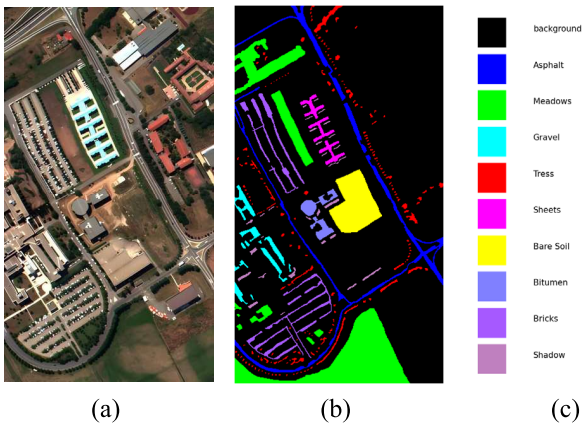


Fig. 8. UP dataset: (a) pseudo-color image, (b) ground-truth map, and (c) category of UP.

with the Adam optimizer. The initial learning rate was set to 0.001 and the model was trained for 5000 iterations on both source and target data. Input cubes were extracted from 7×7 spatial neighborhoods and all convolutional layers were initialized with Xavier normalization. During meta training we adopted episodic learning: for each N-way K-shot task, the number of classes in the support and query sets matched the target dataset (e.g., nine classes for UP, 16 for IP), with $K = 1$ support sample and 19 query samples per class. Theoretically,

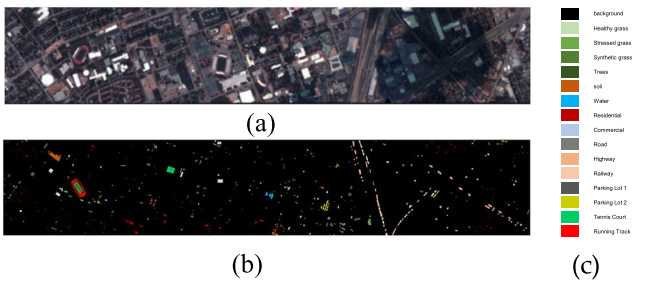


Fig. 9. HT dataset: (a) pseudo-color image, (b) ground-truth map, and (c) category of HT.

TABLE III
PARAMETER TUNING OF THE MOMENTUM COEFFICIENT γ

Dataset	Momentum coefficient γ				
	0.5	0.6	0.7	0.8	0.9
IP	78.99	80.18	**80.64**	80.43	79.33
SA	92.66	93.05	**93.11**	93.06	92.80
UP	82.97	83.42	**83.70**	83.35	82.68
HT	79.56	79.77	**79.96**	79.40	79.83

TABLE IV
PARAMETER TUNING OF THE LOSS WEIGHT λ

Dataset	Loss weight λ				
	0.3	0.4	0.5	0.6	0.7
IP	79.96	80.23	**80.64**	80.58	80.46
SA	93.02	92.95	93.11	**93.17**	93.11
UP	83.50	83.45	83.70	**83.75**	83.60
HT	79.67	79.75	**79.96**	79.68	79.72

TABLE V
PARAMETER TUNING OF THE NOISE WEIGHT η

Dataset	Noise weight η				
	0.6	0.8	1	1.2	1.4
IP	79.98	80.51	80.64	**80.66**	80.51
SA	93.08	93.10	**93.17**	93.07	93.15
UP	83.74	83.10	83.75	83.30	**84.10**
HT	79.58	79.82	**79.96**	79.60	79.76

increasing query samples per class improves generalization and stabilizes training [57]. At test time, five labeled samples per class in the target domain were used to fit a KNN classifier that evaluated the test set. To eliminate random sampling bias, all runs were repeated with ten fixed random seeds, and we report the mean and standard deviation of overall accuracy (OA), average accuracy, and Kappa coefficient.

C. Parameter Tuning

To optimize MEDPL performance across four datasets, we conducted a sensitivity analysis on the core hyperparameters: momentum factor γ, loss weight λ, and the noise weight η. These parameters were selected from the ranges {0.5, 0.6, 0.7, 0.8, 0.9}, {0.3, 0.4, 0.5, 0.6, 0.7}, and {0.6, 0.8, 1, 1.2, 1.4}, respectively. Each run was repeated ten times and the

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TABLE VI
CLASSIFICATION RESULTS (%) ON THE IP DATASET (5 SHOT)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADA-FSL	MLRL	FDFSL	Ours
1	81.95	85.04	99.01	97.80	93.41	97.80	98.29	95.37	**99.02**
2	32.48	31.39	42.91	45.14	47.98	52.18	48.73	53.89	**72.21**
3	36.41	30.22	58.84	46.80	49.35	60.16	62.33	64.08	**66.69**
4	56.59	37.82	81.49	75.60	76.51	86.77	88.62	83.02	**97.20**
5	67.01	62.50	78.63	67.74	66.65	78.70	78.56	74.12	**79.96**
6	65.39	62.91	95.75	86.50	84.14	86.03	88.99	87.24	**97.26**
7	91.74	92.23	99.55	99.57	99.57	99.57	99.57	99.57	**100.00**
8	65.35	59.86	76.90	76.03	83.76	85.31	82.03	86.77	**95.79**
9	80.00	95.24	**100**	**100**	**100**	**100**	**100**	**100**	**100**
10	36.71	50.78	60.51	57.08	58.23	69.71	67.81	63.02	**77.48**
11	41.56	51.70	51.14	61.45	61.32	58.66	61.37	63.83	**72.53**
12	27.93	27.27	62.59	48.69	45.83	81.04	52.82	54.69	**82.43**
13	86.55	91.34	99.29	95.35	96.45	97.65	98.90	98.95	**99.65**
14	62.96	59.24	87.98	89.48	87.83	84.10	88.44	87.75	**92.18**
15	29.19	46.15	75.21	67.72	70.21	79.50	74.91	76.48	**84.44**
16	89.63	81.08	97.70	96.82	96.0	98.30	**99.77**	97.95	98.07
OA	46.77 ±2.94	49.02 ±4.18	65.72 ±3.77	64.89 ±3.50	65.44 ±2.23	68.61 ±3.54	69.38 ±3.87	70.20 ±2.66	**80.66 ±1.38**
AA	59.45 ±1.93	60.30 ±2.30	79.22 ±1.61	75.74 ±2.44	76.08 ±1.48	80.34 ±1.96	80.70 ±1.22	80.42 ±1.95	**88.43 ±1.27**
Kappa	40.55 ±2.95	42.63 ±4.28	61.72 ±3.90	60.36 ±3.96	61.02 ±2.53	64.76 ±3.78	65.70 ±4.11	66.48 ±2.94	**78.14 ±1.54**

mean OA was reported to mitigate randomness. Owing to the high computational cost of exhaustive search, we adopted a suboptimal tuning strategy; the results are presented in Tables III–V. The results indicate that for the IP, SA, UP, and HT datasets, the momentum factor γ consistently achieved optimal performance at 0.7. The loss weight λ yielded optimal values of 0.5, 0.6, 0.6, and 0.5 for the respective datasets, while the noise weight η achieved optimal values of 1.2, 1, 1.4, and 1.

The momentum factor γ controls the decay rate of historical prototype weights. As γ increases from 0.5 to 0.9, performance first rises and then falls, peaking at 0.7. This indicates that moderate retention of historical prototypes effectively suppresses class-center drift caused by random sampling noise, whereas excessive reliance on past information hampers adaptation to new tasks. Similarly, λ balances the contribution of the two prototype losses; performance is maximized when the losses are roughly equal, whereas extreme values of λ degrade accuracy, confirming the necessity of the cooperative dual-prototype mechanism. Finally, the noise intensity coefficient η controls the magnitude of injected noise. When η increases from 0.6, moderate and controllable noise effectively enhances model generalization by increasing feature diversity. However, once η exceeds specific thresholds, excessive noise corrupts discriminative information in the original features, leading to performance degradation.

D. Quantitative Comparisons

To validate the effectiveness of MEDPL in few-shot hyperspectral scenarios, we compare it with eight baselines: the traditional approach SVM [17], two deep methods (3DCNN [20] and SSRN [21]), and five metric-learning-based few-shot methods (DCFSL [36], Gia-FSL [37], ADAFSL [38], MLRL [58], and FDFSL [39]). Tables VI–IX present 5-shot classification results on IP, SA, UP, and HT. MEDPL achieves state-of-the-art performance on all four datasets, demonstrating significant advantages over the suboptimal method FDFSL. On the IP dataset, MEDPL outperforms FDFSL by 10.46% in OA, 8.01% in AA, and 11.66% in Kappa. On the SA dataset, improvements reach 2.32% in OA, 2.01% in AA, and 2.58% in Kappa. On the UP dataset, MEDPL exceeds FDFSL by 0.79% in OA, 5.18% in AA, and 0.98% in Kappa. Finally, on the HT dataset, the advantages are 1.39% in OA, 1.30% in AA, and 1.52% in Kappa.

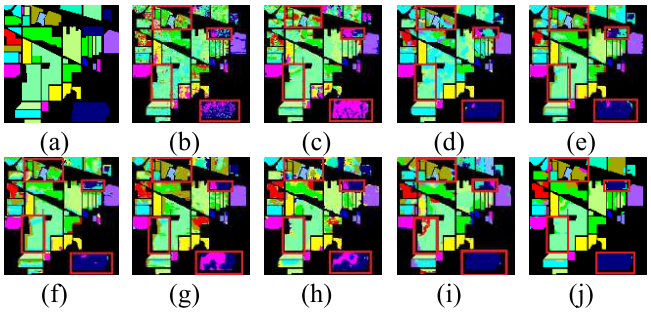


Fig. 10. Visualization results of various methods on the IP dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) Gia-CFSL, (g) ADAFSL, (h) MLRL, (i) FDFSL, and (j) MEDPL.

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TABLE VII
CLASSIFICATION RESULTS (%) ON THE SA DATASET (5 SHOT)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADA-FSL	MLRL	FDFSL	Ours
1	96.02	93.08	98.53	99.45	98.09	99.63	**99.97**	98.86	99.62
2	94.71	95.99	99.67	98.82	99.40	99.96	98.75	99.27	**100.00**
3	88.56	89.80	96.87	97.31	95.46	**97.75**	88.73	91.21	97.07
4	98.83	98.15	99.31	99.26	98.52	**99.83**	99.21	98.91	99.01
5	94.01	92.34	95.79	91.52	86.55	96.20	91.12	94.13	**96.86**
6	98.39	98.80	99.66	99.84	99.25	99.84	99.59	98.97	**99.94**
7	96.61	97.83	**99.93**	99.69	99.28	98.75	99.59	99.08	99.81
8	56.11	59.31	66.17	79.48	81.31	77.51	80.63	79.37	**82.12**
9	94.04	96.93	99.79	99.32	97.76	99.52	99.69	98.59	**99.87**
10	76.44	78.10	90.73	86.78	83.73	85.93	85.83	85.50	**92.30**
11	91.06	89.85	97.34	97.01	93.56	**99.72**	97.66	96.39	99.42
12	99.29	99.34	99.88	99.17	99.19	99.83	98.73	99.39	**99.94**
13	95.51	95.17	99.01	97.85	94.27	**99.49**	98.49	97.67	98.76
14	91.87	91.24	98.13	98.43	98.46	98.56	98.15	**99.52**	97.77
15	64.62	59.06	81.47	75.73	74.51	75.05	72.62	81.23	**84.86**
16	84.08	83.21	94.43	92.75	90.32	**97.21**	92.33	91.54	94.40
OA	81.42 ±2.99	81.74 ±2.31	89.13 ±2.59	90.51 ±0.99	89.75 ±0.80	90.52 ±1.90	89.97 ±1.73	90.85 ±0.90	**93.17 ±0.48**
AA	88.76 ±1.34	88.64 ±1.57	94.79 ±0.55	94.52 ±0.66	93.10 ±1.19	95.30 ±0.99	93.82 ±1.54	94.35 ±0.98	**96.36 ±0.66**
Kappa	79.44 ±3.23	79.72 ±2.53	87.95 ±2.80	89.45 ±1.09	88.60 ±0.90	89.48 ±2.09	88.83 ±1.92	89.83 ±0.99	**92.41 ±0.54**

TABLE VIII
CLASSIFICATION RESULTS (%) ON THE UP DATASET (5 SHOT)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADA-FSL	MLRL	FDFSL	Ours
1	88.98	59.82	**91.84**	78.09	73.25	80.56	77.65	86.07	90.58
2	83.91	63.05	**95.13**	81.63	84.95	81.72	81.39	83.09	76.39
3	39.98	68.91	55.23	64.72	62.70	72.09	61.55	65.16	**85.23**
4	60.22	77.31	78.02	**94.65**	94.05	87.90	87.81	89.29	90.10
5	95.44	90.77	98.34	99.38	99.65	98.45	99.81	99.40	**99.90**
6	37.12	63.40	53.56	74.94	73.73	73.70	69.49	75.11	**85.99**
7	40.62	87.64	60.07	78.44	81.94	87.05	82.48	88.66	**97.76**
8	68.17	57.27	85.34	68.25	72.34	76.84	70.15	85.31	**88.92**
9	**99.13**	95.57	98.08	98.62	96.18	99.11	96.88	94.62	98.47
OA	64.12 ±4.55	65.74 ±1.77	76.26 ±5.78	80.08 ±3.92	80.91 ±1.29	81.22 ±4.52	78.88 ±3.65	83.31 ±1.89	**84.10 ±3.47**
AA	68.18 ±2.27	73.72 ±1.01	79.51 ±3.21	82.08 ±1.91	82.09 ±2.04	84.16 ±2.57	80.80 ±2.11	85.19 ±2.23	**90.37 ±1.54**
Kappa	55.59 ±4.64	57.37 ±1.97	70.56 ±6.69	74.46 ±4.45	75.33 ±1.53	75.82 ±5.15	72.80 ±4.19	78.84 ±2.16	**79.82 ±4.07**

Among these methods, the traditional classifier SVM yields the lowest classification accuracy due to its inability to leverage spatial information effectively and poor performance under high-dimensional, few-sample constraints. In contrast, deep convolutional networks like 3DCNN and SSRN jointly extract spatial–spectral features to model complex hyperspectral patterns. However, under sample scarcity, these models suffer from overfitting and inadequate training, resulting in

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TABLE IX
CLASSIFICATION RESULTS (%) ON THE HT DATASET (5 SHOT)

Class	SVM	3DCNN	SSRN	DCFSL	Gia-CFSL	ADA-FSL	MLRL	FDFSL	Ours
1	81.03	89.24	81.94	**90.20**	89.66	89.56	88.68	87.38	86.94
2	74.54	77.15	80.72	74.45	74.43	87.56	83.75	**88.06**	86.77
3	97.60	95.06	99.68	94.65	93.48	97.33	96.40	97.90	**99.74**
4	86.28	88.15	89.81	91.38	89.18	89.96	92.12	92.07	**92.87**
5	89.06	94.68	95.78	96.90	94.69	95.73	94.77	**99.42**	97.32
6	82.52	72.12	83.57	84.34	79.61	78.34	82.44	85.06	**88.62**
7	55.96	54.53	66.79	59.29	58.38	60.84	73.50	69.70	**80.02**
8	24.85	42.38	22.36	42.59	37.74	42.24	42.18	**48.42**	39.59
9	58.49	59.98	61.50	59.95	55.93	56.12	68.91	**73.17**	72.03
10	47.67	53.29	49.14	66.79	60.37	**68.54**	62.63	64.97	67.97
11	51.04	61.13	**77.97**	56.02	59.46	65.55	60.33	64.52	77.11
12	39.67	44.77	40.70	**75.42**	71.84	71.87	65.18	62.14	64.32
13	19.61	31.56	75.09	86.73	88.64	44.94	91.25	**93.00**	90.00
14	93.48	86.36	**99.80**	88.60	87.42	94.52	89.67	98.16	99.67
15	96.05	96.56	99.77	94.65	91.41	94.99	97.05	99.39	**100.00**
OA	63.75 ±2.26	68.58 ±2.58	70.46 ±1.24	74.22 ±2.92	72.26 ±3.60	74.77 ±2.85	76.63 ±2.28	78.57 ±1.98	**79.96 ±2.12**
AA	66.52 ±1.91	69.80 ±2.61	74.97 ±0.99	77.47 ±2.62	75.48 ±2.82	75.87 ±3.01	79.30 ±2.10	81.56 ±1.55	**82.86 ±1.75**
Kappa	60.89 ±2.39	65.92 ±2.76	68.08 ±1.33	72.16 ±3.16	70.05 ±3.89	72.72 ±3.08	74.75 ±2.47	76.86 ±2.12	**78.38 ±2.28**

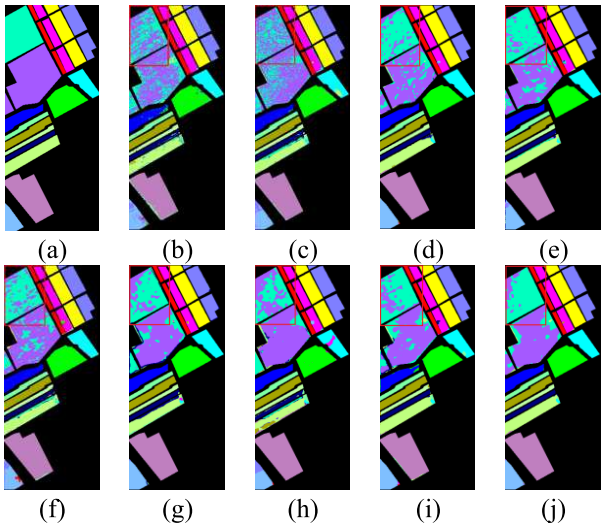


Fig. 11. Visualization results of various methods on the SA dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) Gia-CFSL, (g) ADAFSL, (h) MLRL, (i) FDFSL, and (j) MEDPL.

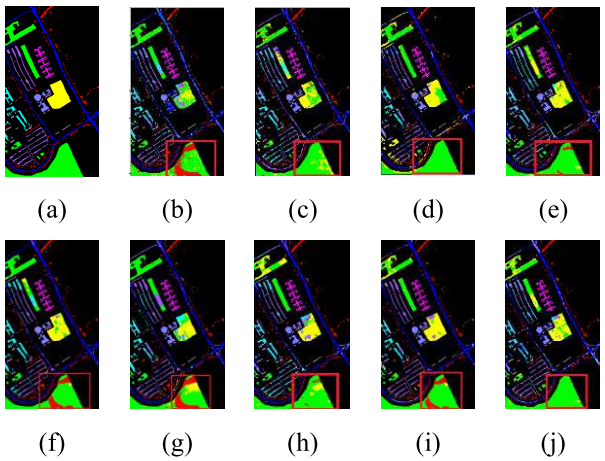


Fig. 12. Visualization results of various methods on the UP dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) Gia-CFSL, (g) ADAFSL, (h) MLRL, (i) FDFSL, and (j) MEDPL.

limited performance gains. Prototype-based few-shot methods, including DCFSL, Gia-FSL, and ADAFSL, employ domain adaptation to align feature distributions between source and target domains, enhancing target-domain generalization. The MLRL method further integrates contrastive learning to reduce intraclass variance while utilizing cross-attention modules to strengthen query-support sample relationships, allowing queries to focus on class-relevant supports and refining prototype classification. Meanwhile, FDFSL enhances target-domain prototype robustness through feature disentanglement that suppresses source-domain bias. Ultimately, MEDPL's dual-prototype metric learning framework, combining accumulated and enhanced prototypes, simultaneously elevates classification accuracy and generalization capability, achieving optimal performance across all evaluated methods.

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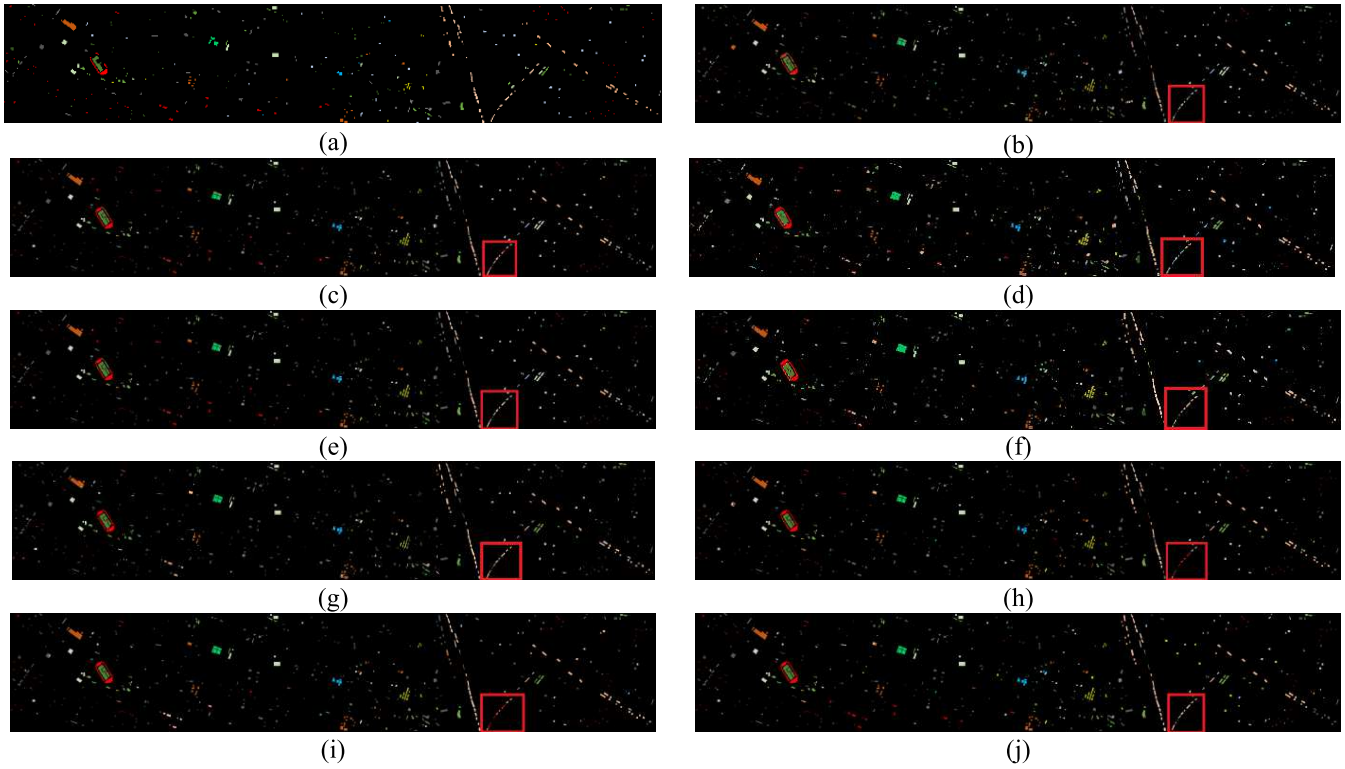


Fig. 13. Visualization results of various methods on the HT dataset. (a) Ground-truth, (b) SVM, (c) 3DCNN, (d) SSRN, (e) DCFSL, (f) Gia-CFSL, (g) ADAFSL, (h) MLRL, (i) FDFSL, and (j) MEDPL.

E. Qualitative Comparisons

Figs. 10–13 visualize the classification maps of all the compared methods and the MEDPL method on the IP, SA, UP, and HT datasets. Among them, the classification maps generated by MEDPL are the closest to the ground-truth maps, especially on the IP dataset. The IP dataset is characterized by an imbalanced sample distribution and high noise, which makes it difficult for the previous few-shot methods to accurately distinguish complex boundaries. However, the improvements in model accuracy and generalization brought by MEDPL enhance the robustness of the model, thereby effectively improving the classification results on the IP dataset. This is particularly evident in the second class "corn notill," the sixth class "grass trees in farmland," the tenth class "soybean notill," and the twelfth class "soybean clean." The SA dataset has a high signal-to-noise ratio and homogeneous scene characteristics. Among the deep learning methods, SSRN performs relatively well on this dataset, and the few-shot methods also show significant improvements in classification results. MEDPL further improves the results for the tenth class "corn senesced green weeds" and the fifteenth class "vineyard untrained." The high spatial resolution of the UP dataset reduces the mixed-pixel effect, and the few-shot methods achieve relatively significant classification results on this dataset. Compared with other methods, MEDPL shows improvements in the third class "gravel," the sixth class "bare soil," and the seventh class "bitumen." In the HT dataset, the classes are scattered and the boundaries are clear. The classification map generated by MEDPL for the eleventh class "railway" is the closest to the ground-truth map. In summary, the visualization of the classification maps further highlights the robustness of the MEDPL method in complex scenarios.

TABLE X
ABLATION COMPARISON RESULTS (%) OF MEDPL

Module			Dataset			
FSL	MU	PA	IP	SA	UP	HT
√			75.49	92.03	81.94	78.20
√	√		75.87	93.07	82.54	79.57
√		√	77.40	92.47	82.68	79.46
√	√	√	**80.66**	**93.17**	**84.10**	**79.96**

To comprehensively evaluate the classification performance of MEDPL across different sample sizes, a systematic comparison with baseline methods was conducted under 1-shot to 5-shot settings, with results shown in Fig. 14. It can be observed that on all four datasets, as the number of samples per class increases from 1 to 5, the OA of all baseline methods demonstrates an upward trend, consistent with the fundamental principles of FSL. Meanwhile, the MEDPL method consistently outperforms other baseline approaches across all sample settings. Specifically, MEDPL's advantage is particularly prominent under extremely low-sample conditions (e.g., 1-shot and 3-shot). In the 1-shot setting, MEDPL already achieves significantly higher OA than other methods. This demonstrates that the dual-prototype mechanism can more efficiently utilize limited sample information under extreme scarcity, enhancing both the discriminative power and generalization capability of the model while preventing overfitting to limited samples.

In addition, to intuitively compare the feature extraction effects of the latest MLRL and FDFSL methods and the

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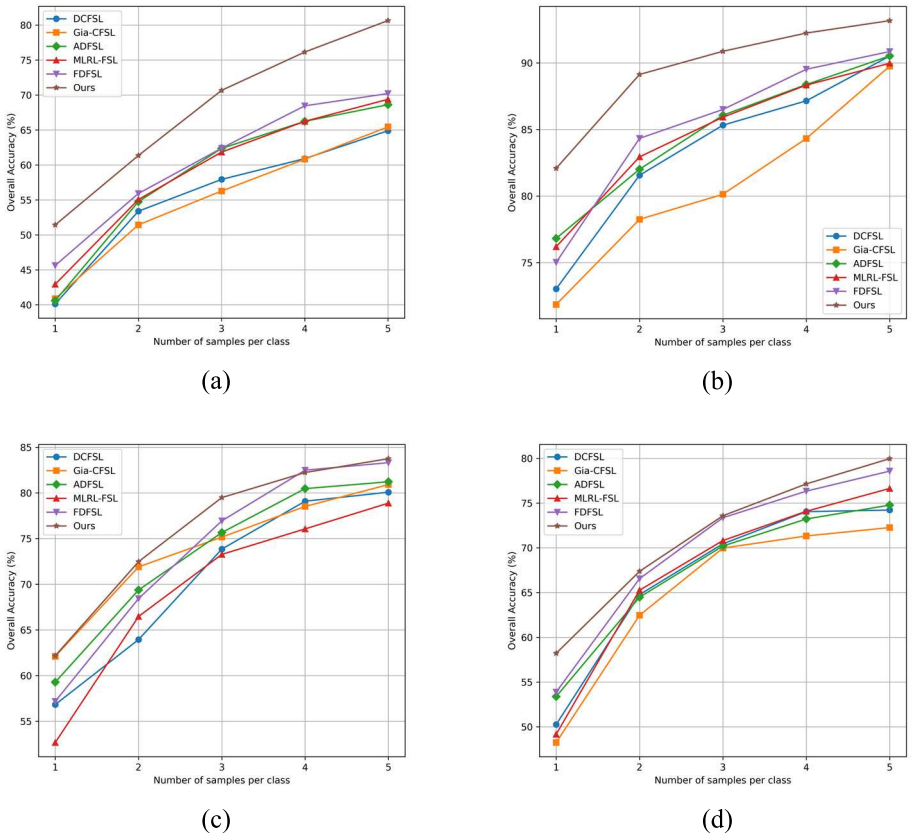


Fig. 14. OA (%) of all methods from 1- to 5-shot settings: (a) IP, (b) SA, (c) UP, and (d) HT.

TABLE XI
TRAINING TIME (S), TESTING TIME (S), AND PARAMETERS (M) ON EACH DATASET WITH ALL METHODS

Dataset	Metrics	Method					
		DCFSL	Gia-CFSL	ADAFSL	MLRL	FDFSL	MEDPL
IP	Training time	2101.45s	4269.32s	2501.45s	3633.43s	2453.21s	1861.30s
	Testing time	5.67s	2.56s	6.53s	1.73s	5.79s	0.64s
	#Params	4.28M	0.86M	3.21M	6.07M	5.28M	0.19M
SA	Training time	2760.51s	4252.35s	3120.12s	3636.96s	3136.15s	1326.05s
	Testing time	23.54s	10.25s	21.25s	8.97s	26.78s	2.43s
	#Params	4.28M	0.86M	3.21M	6.08M	5.28M	0.19M
UP	Training time	1534.98s	4001.86s	2980s	2843.12s	2857s	1596.84s
	Testing time	6.84s	5.33s	19.23s	5.50s	15.43s	2.77s
	#Params	4.25M	0.86M	3.21M	6.04M	5.28M	0.18M
HT	Training time	1856.23s	3253.28s	2563.4s	2438.25s	1832.52s	1450.98s
	Testing time	4.68s	1.98s	5.17s	3.65s	4.32s	0.74s
	#Params	4.28M	0.86M	3.21M	6.08M	5.28M	0.19M

proposed MEDPL method on the four datasets, Fig. 15 uses t-SNE technology to show the 2-D feature distributions of samples from different classes in the IP, SA, UP, and HT datasets. It is evident that MEDPL outperforms MLRL and FDFSL in terms of interclass separability and intraclass compactness, with clearer classification boundaries. Specifically, as shown in Fig. 15(a)–(c), the comparison methods exhibit intermingling among classes 2, 3, 11, and 12 in the IP dataset, while the proposed method achieves stronger intraclass clustering and higher interclass separation for these four classes in the feature space. In Fig. 15(d)–(f), the comparison methods show overlap between classes 8 and 15 in the SA dataset, whereas the proposed method alleviates this overlap. In Fig. 15(g)–(i), the comparison methods display pairwise overlap between

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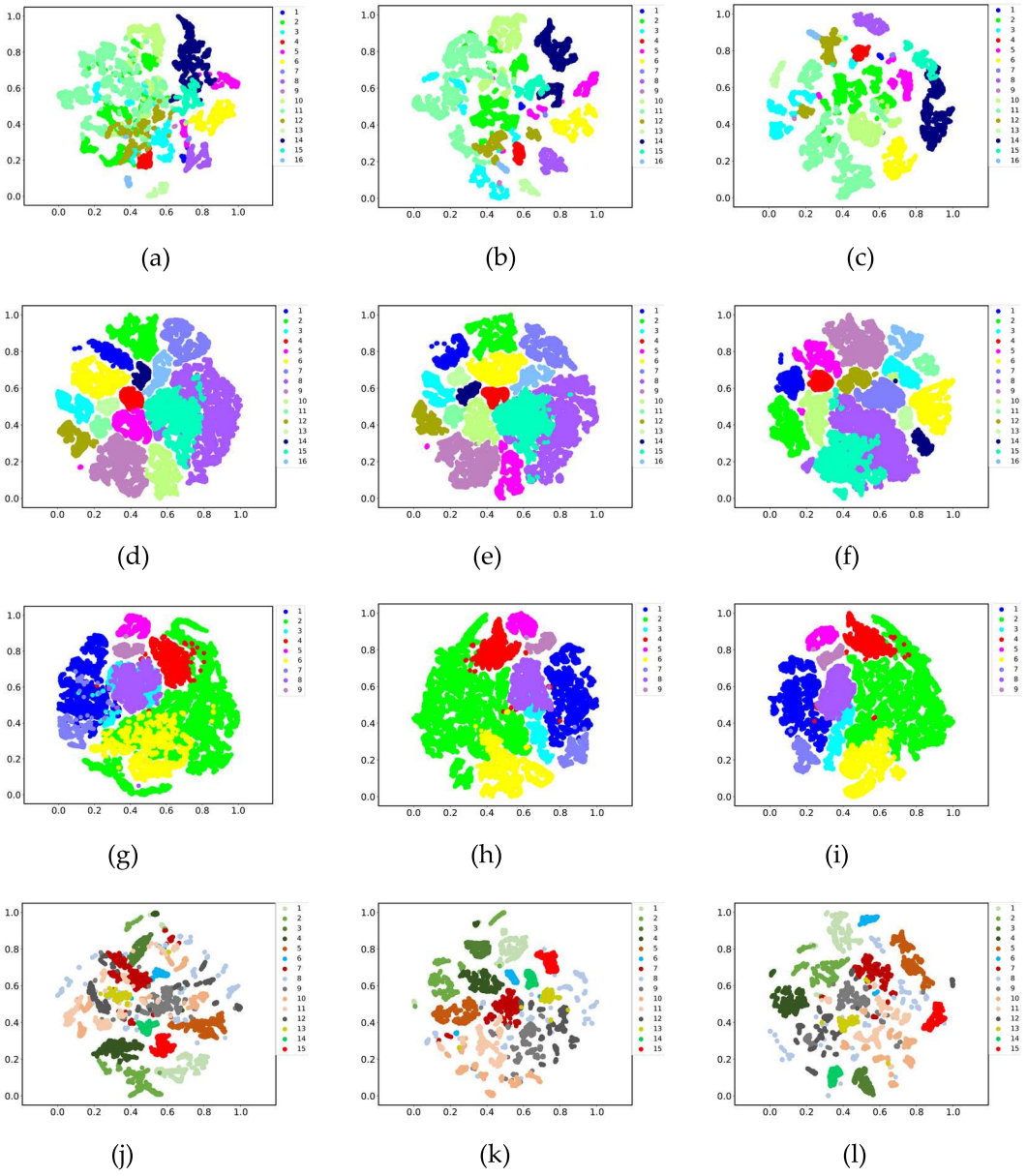


Fig. 15. 2-D feature visualization on four datasets. (a) MLRL in IP. (b) FDFSL in IP. (c) MEDPL in IP. (d) MLRL in SA. (e) FDFSL in SA. (f) MEDPL in SA. (g) MLRL in UP. (h) FDFSL in UP. (i) MEDPL in UP. (j) MLRL in HT. (k) FDFSL in HT. (l) MEDPL in HT.

classes 2 and 6, and classes 3 and 8 in the UP dataset, while the proposed method clearly distinguishes these overlapping classes. In Fig. 15(j)–(l), the proposed method achieves the highest degree of separation between classes in the HT dataset.

F. Ablation Experiments

The core of the proposed MEDPL method lies in the momentum update module and the perturbation augmentation module. To verify the effectiveness of these two modules, ablation studies were conducted to assess their individual contributions, primarily investigating four model variants: 1) FSL with only the ordinary mean prototype in both the source and target domains under the current task; 2) adding the momentum update to 1); 3) adding the perturbation augmentation to 1); and 4) combining both the momentum update and perturbation augmentation with 1).

As shown in Table X, model 2) achieved an increase in the OA metric of 0.38%, 1.04%, 0.6%, and 1.37% on the IP, SA, UP, and HT datasets, respectively, compared to model 1), demonstrating that the momentum-updated accumulated prototype is a more effective class representation. Model 3) saw an increase in the OA metric of 1.91%, 0.44%, 0.74%, and 1.26% on the IP, SA, UP, and HT datasets, respectively, compared to model 1), proving that the perturbation-augmented prototype enhances the generalizability of the class representation. This moderate and controlled class-conditional noise not only preserves valid feature representations but also acts as an effective regularization mechanism, significantly enhancing model generalization capability. Model 4) achieved an increase in the OA metric of 4.79%, 0.1%, 1.56%, and 0.39% on the IP, SA, UP, and HT datasets, respectively, compared to model 2), and an increase of 3.26%, 0.7%, 1.42%, and

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0.5% compared to model 3). This demonstrates that the dual-prototype metric learning mechanism improves the model's accuracy and robustness across different tasks.

G. Computational Complexity Analysis

To demonstrate the computational complexity of different few-shot HSI classification methods, Table XI lists the training time, testing time, and model parameter counts for the DCFSL, Gia-CFSL, ADAFSL, MLRL, FDFSL, and MEDPL methods across the four datasets. It can be seen that the proposed MEDPL method has the shortest training time, the least testing time consumption, and a minimal number of parameters among these few-shot methods. Since few-shot methods rely on source-domain data to assist in training, their training times are relatively long. The proposed MEDPL method, which constructs the model using only a small number of parameters in the convolutional layers, significantly reduces the model's parameter count. This, in turn, shortens the training duration and reduces the resource consumption required for training.

V. Conclusion

This article proposes a novel MEDPL method to achieve robust few-shot HSI classification. The core of this method lies in the momentum-updating mechanism and the class-conditional perturbation-augmentation strategy. Specifically, by accumulating historical prototype information, it effectively suppresses the class center offset caused by random sampling noise and sample bias, significantly enhancing the stability of prototype representation. Meanwhile, by adding noise perturbations that conform to the class-conditional distribution characteristics to the support set features, it expands the intra-class feature diversity, thereby strengthening the generalization capability of the prototype. Finally, through dual-prototype metric learning that combines accumulated prototypes and enhanced prototypes, it collaboratively improves the model's discriminative ability for specific tasks and its robustness across tasks. Extensive experimental results on the IP, SA, UP, and HT datasets demonstrate that the proposed method can significantly enhance the classification performance of the model compared with existing FSL methods.

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Hanchi Liu received the B.E. degree in data science and big data technology from the School of Mathematics and Computer Science, Yan'an University, Yan'an, China, in 2022, where he is currently pursuing the master's degree in electronic information.

His research interests include computer vision, few-shot learning, and object detection.



Jinrong He (Member, IEEE) received the B.S. degree in science of information and computation and the M.S. degree in computational mathematics from Wuhan University of Technology, Wuhan, China, in 2007 and 2010, respectively, and the Ph.D. degree in computer software and theory from the State Key Laboratory of Software Engineering, Wuhan University, Wuhan, in 2014.

From 2015 to 2018, he served as a Lecturer at Northwest A&F University, Xianyang, China. Currently, he is an Associate Professor with Yan'an University, Yan'an, China. His research interests include computer vision, machine learning, and few-shot learning.



Xiangqing Zhang received the Ph.D. degree from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2024.

He is currently a Lecturer with the College of Mathematics and Computer Science, Yan'an University, Yan'an, China. His research interests include deep learning, data mining, and object detection of remote sensing images.



Zhaokui Li (Member, IEEE) received the M.S. degree in computer application from Liaoning University, Shenyang, China, in 2003, and the Ph.D. degree in computer software and theory from Wuhan University, Wuhan, China, in 2014.

He is currently a Professor with the School of Computer, Shenyang Aerospace University, Shenyang. His research interests include hyperspectral image analysis, computer vision, and machine learning.

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